B4 Notes: Powers, roots and exponential growth

← B4 overview

This module starts with something you already know — $2^3 = 2 \cdot 2 \cdot 2$ — and asks a bold question: what could 2^0, 2^{-3} or $2^{1/2}$ possibly mean? The answer is not "whatever the teacher says". It is forced on us by one simple wish: *the rules must keep working*. Once powers make sense for all these exponents, we use them to describe things that grow or shrink by the same factor again and again: bacteria, savings, medicine in your blood, the value of a car. We end with graphs of power functions and a first look at things that repeat, like tides and Ferris wheels.

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1. Rules for powers

In a^n the number a is the base (*grondtal*) and n is the exponent (*exponent*). For a whole number $n \geq 1$,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}.$$

Everything in this section comes from **counting factors**.

Rule 1 (same base, multiply). $a^p \cdot a^q = a^{p+q}$. $a^3 \cdot a^2 = (a \cdot a \cdot a)(a \cdot a)$: three factors and then two more, five in total, so a^5.

Rule 2 (same base, divide). $\dfrac{a^p}{a^q} = a^{p-q}$ for $p > q$ and $a \neq 0$. $\dfrac{a^5}{a^2} = \dfrac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a}$: two factors cancel, three remain, so a^3.

Rule 3 (power of a power). $(a^p)^q = a^{pq}$. $(a^2)^3 = a^2 \cdot a^2 \cdot a^2$: three groups of two factors, six in total.

Rule 4 (power of a product). $(ab)^p = a^p b^p$. $(ab)^3 = (ab)(ab)(ab) = (aaa)(bbb)$, because multiplication may be done in any order.

Together these are the *rekenregels voor machten*.

Worked example 1. Simplify $(2x^3)^4 \cdot 3x^2$. By rule 4, $(2x^3)^4 = 2^4(x^3)^4 = 16x^{12}$ (rule 3). Then $16x^{12} \cdot 3x^2 = 48x^{14}$ (rule 1).

Worked example 2. Write $8 \cdot 4^5$ as a single power of 2. $8 = 2^3$ and $4^5 = (2^2)^5 = 2^{10}$, so $8 \cdot 4^5 = 2^{13}$.

Common mistakes

- $(-2)^2 = 4$ but $-2^2 = -4$. Powers are calculated before the minus sign, so -2^2 means $-(2^2)$. Only brackets make the minus part of the base.
- $2^3 \cdot 3^2 \neq 6^5$. The rules only combine powers with the **same base** (rules 1–3) or the **same exponent** (rule 4). Here $2^3 \cdot 3^2 = 8 \cdot 9 = 72$, while $6^5 = 7776$.
- $2^3 + 2^4 \neq 2^7$. There is no rule for sums. You can factor, though: $2^3 + 2^4 = 2^3(1 + 2) = 24$.
- $(a + b)^2 \neq a^2 + b^2$. Rule 4 is about products, not sums. You met $(a + b)^2 = a^2 + 2ab + b^2$ in B2.
- $(a^3)^2 = a^6$, not a^9; and $a^3 \cdot a^2 = a^5$, not a^6.

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2. Zero and negative exponents

"a^0 = zero factors of a" means nothing. So we have a free choice — but only one choice keeps rule 2 alive. Rule 2 says $\dfrac{a^3}{a^3} = a^{3-3} = a^0$. But anything (non-zero) divided by itself is 1. So if the rule is to survive, we **must** define

$$a^0 = 1 \quad (a \neq 0).$$

In the same way, $\dfrac{a^2}{a^5}$ is $\dfrac{1}{a^3}$ after cancelling, and rule 2 would call it $a^{2-5} = a^{-3}$. So we **must** define

$$a^{-n} = \frac{1}{a^n} \quad (a \neq 0).$$

Mathematicians call this idea the *principle of permanence*: when you extend a definition to new cases, choose the extension that keeps the old rules true. It is the same reason that $(-1)\cdot(-1) = 1$.

You can also see it in a pattern. Each step to the left divides by 2:

n	3	2	1	0	-1	-2	-3
2^n	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

Now check that all four rules really hold for every whole exponent, positive, zero or negative. For example $2^{-3}\cdot 2^5 = \frac{1}{8}\cdot 32 = 4 = 2^2 = 2^{-3+5}$. They do, and you will never again need the condition $p > q$ in rule 2.

Worked example 3. $\left(\frac{2}{3}\right)^{-2} = \dfrac{1}{(2/3)^2} = \dfrac{1}{4/9} = \dfrac{9}{4}$. A useful shortcut: $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^{n}$.

Common mistake. 2^{-3} is **not** negative. It is $\frac{1}{8}$. A negative exponent means "divide", not "make negative".

> **Going further: what is 0^0?** Rule 2 cannot help, because $\frac{0^3}{0^3}$ is $\frac{0}{0}$, which is undefined. Two patterns disagree: $x^0 = 1$ for every $x \neq 0$ suggests $0^0 = 1$, while $0^x = 0$ for every $x > 0$ suggests $0^0 = 0$. In algebra and counting, mathematicians usually agree that $0^0 = 1$ (for example, so that $(1+x)^n$ expands neatly). In calculus they say "0^0 is an indeterminate form", meaning that a limit of that shape can come out as anything. So: it is a convention, and it depends on the context.

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3. Scientific notation

Very large and very small numbers are written in scientific notation (*wetenschappelijke notatie*): a number from 1 up to (not including) 10, times a power of 10.

$$384\,400\,000 = 3.844 \cdot 10^8, \qquad 0.000\,0052 = 5.2 \cdot 10^{-6}.$$

The exponent counts how many places the decimal point moves. Negative exponents (§2) are exactly what makes small numbers fit the same pattern.

Worked example 4. $(4 \cdot 10^7) \cdot (5 \cdot 10^{-3}) = 20 \cdot 10^4 = 2.0 \cdot 10^5$. Multiply the front numbers, add the exponents (rule 1), then tidy up so the front number is between 1 and 10.

On a calculator, $3.2 \cdot 10^{-4}$ is entered as `3.2 E -4` or `3.2 ×10^x -4`.

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4. Square roots

For $a \geq 0$, $\sqrt{a}$ is the number ≥ 0 whose square is a. So $\sqrt{9} = 3$, not ± 3. The equation $x^2 = 9$ has two solutions, $x = 3$ and $x = -3$, but the symbol $\sqrt{9}$ means only the non-negative one.

Product rule. For $a, b \geq 0$:

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}.$$

Why: both sides are ≥ 0. Square the left side: $(\sqrt{a}\sqrt{b})^2 = (\sqrt{a})^2(\sqrt{b})^2 = ab$ (rule 4). So $\sqrt{a}\sqrt{b}$ is a non-negative number whose square is ab, and that is exactly the definition of $\sqrt{ab}$. In the same way $\dfrac{\sqrt{a}}{\sqrt{b}} = \sqrt{\dfrac{a}{b}}$ for $a \geq 0$, $b > 0$.

Simplifying roots (*wortels vereenvoudigen*). Look for the largest square factor:

$$\sqrt{12} = \sqrt{4 \cdot 3} = \sqrt{4} \cdot \sqrt{3} = 2\sqrt{3}, \qquad \sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}.$$

Simplified roots can be added like "like terms": $\sqrt{18} + \sqrt{8} = 3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2}$.

There is no sum rule. $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ in general. Test: $\sqrt{9+16} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. *Why it fails:* $(\sqrt{a} + \sqrt{b})^2 = a + 2\sqrt{ab} + b$, which is bigger than $a + b$ unless a or b is 0. Geometrically: walking along the two legs of a right triangle (3 and 4) is longer than walking along the hypotenuse (5).

Rationalising the denominator (*wortel uit de noemer halen*). Multiply top and bottom by the root, which does not change the value because you multiply by $\frac{\sqrt{3}}{\sqrt{3}} = 1$:

$$\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}.$$

Why bother? Before calculators, $\frac{1}{\sqrt{2}} = \frac{1}{1.41421\ldots}$ was a horrible long division, while $\frac{\sqrt{2}}{2} = 0.70710\ldots$ is easy. Today the main reason is that answers in the same form are easy to compare, and Dutch tests expect it.

Worked example 5 (conjugate). For $\dfrac{2}{\sqrt{3}-1}$, multiply by $\dfrac{\sqrt{3}+1}{\sqrt{3}+1}$. The denominator becomes $(\sqrt{3})^2 - 1^2 = 2$ (the product $(a-b)(a+b) = a^2 - b^2$ from B2), so the fraction equals $\dfrac{2(\sqrt{3}+1)}{2} = \sqrt{3}+1$.

Common mistakes

- $\sqrt{x^2} = |x|$, not x: $\sqrt{(-5)^2} = \sqrt{25} = 5$.
- $2\sqrt{3} + 3\sqrt{2}$ cannot be combined: the roots are different.
- $\sqrt{4 + x^2}$ does **not** simplify to $2 + x$.

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5. n-th roots and fractional exponents

$\sqrt[3]{a}$ (the cube root) is the number whose cube is a: $\sqrt[3]{8} = 2$ and $\sqrt[3]{-8} = -2$. In general $\sqrt[n]{a}$ is the number whose n-th power is a. For even n we need $a \geq 0$ and take the non-negative answer; for odd n every real a is allowed.

Now the permanence idea again. What should $a^{1/2}$ mean? Rule 3 demands

$$\left(a^{1/2}\right)^2 = a^{\frac{1}{2}\cdot 2} = a^1 = a.$$

So $a^{1/2}$ must be a number whose square is a: we define $a^{1/2} = \sqrt{a}$. Likewise $\left(a^{1/n}\right)^n = a$ forces

$$a^{1/n} = \sqrt[n]{a}, \qquad a^{p/q} = \left(\sqrt[q]{a}\right)^p = \sqrt[q]{a^p} \quad (a > 0).$$

Worked example 6.

- $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$. Take the root first; the numbers stay small.
- $16^{-3/4} = \dfrac{1}{16^{3/4}} = \dfrac{1}{(\sqrt[4]{16})^3} = \dfrac{1}{2^3} = \dfrac{1}{8}$.
- $\sqrt{x} \cdot x^3 = x^{1/2} \cdot x^3 = x^{7/2}$ for $x > 0$.

All four rules from §1 keep holding for fractional exponents as long as the base is **positive**. With a negative base things break: $(-8)^{1/3} = -2$ looks fine, but $(-8)^{2/6}$ would be $\sqrt[6]{(-8)^2} = \sqrt[6]{64} = 2$, even though $\frac{2}{6} = \frac{1}{3}$. That is why the rules for fractional exponents are only used with $a > 0$.

Simplifying cube roots: $\sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$. Simplifying works the same way, now with cube factors.

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6. $\sqrt{2}$ is irrational

A rational number (*rationaal getal*) is a fraction $\frac{p}{q}$ of whole numbers with $q \neq 0$. The Greeks of Pythagoras's school believed every length was rational. The diagonal of a unit square, length $\sqrt{2}$, proved them wrong. Here is the proof. It is a **proof by contradiction**: assume the opposite, and show that leads to nonsense.

Theorem. $\sqrt{2}$ is not rational.

Proof. Suppose $\sqrt{2} = \frac{p}{q}$ with p, q positive whole numbers, and the fraction **in lowest terms** (every fraction can be simplified until p and q have no common factor). Square both sides: $2 = \frac{p^2}{q^2}$, so

$$p^2 = 2q^2.$$

So p^2 is even. Then p itself is even, because the square of an odd number is odd: $(2k+1)^2 = 4k^2 + 4k + 1$. Write $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$, which is even, so q is even too. Now p and q are both even, so the fraction was **not** in lowest terms. Contradiction. So $\sqrt{2}$ is not rational. ■

Consequence: the decimal expansion $1.41421356\ldots$ never ends and never repeats. In problem 24 you will prove the same for $\sqrt{3}$ and see exactly where the argument would break for $\sqrt{4}$.

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7. Exponential growth and decay

Repeated percentages multiply. If an amount grows by 5%, the new amount is $100\% + 5\% = 105\%$ of the old one: you **multiply by 1.05**. After a second year you multiply by 1.05 again, because the 5% is now taken of the bigger amount. So after t years you have multiplied by 1.05^t. This is exponential growth (*exponentiële groei*):

$$N = b \cdot g^t$$

- b = start value (*beginwaarde*), the value at $t = 0$;
- g = growth factor (*groeifactor*) per time unit;
- t = number of time units.

If $g > 1$ the quantity grows; if $0 < g < 1$ it decays (*exponentiële afname*).

Change per period	Growth factor g
increase of 7%	$1 + 0.07 = 1.07$
decrease of 12%	$1 - 0.12 = 0.88$
doubles	2
increase of 150%	2.5
halves	0.5

And back: $g = 1.035$ means +3.5% per period; $g = 0.94$ means -6% per period.

Worked example 7. A savings account starts at €5000 with 4% interest per year. Then $N = 5000 \cdot 1.04^t$. After 3 years: $5000 \cdot 1.04^3 = 5624.32$, so €5624.32.

Common mistakes

- **Adding percentages.** "5% per year for 10 years is 50%" is wrong. Over 10 years the factor is $1.05^{10} \approx 1.6289$, an increase of about 62.9%. Interest is earned on interest.
- **Undoing a percentage.** A 20% decrease followed by a 20% increase gives $0.8 \cdot 1.2 = 0.96$: you lost 4%. The increase is taken of a smaller amount.
- **Wrong start value.** b is the value at $t = 0$, not after the first period.

Going further: compound interest and the number e Suppose a bank gives 100% interest per year on €1. Paid once a year you get $(1 + 1)^1 = 2$. Paid twice a year (50% each half year): $(1 + \frac{1}{2})^2 = 2.25$. Monthly: $(1 + \frac{1}{12})^{12} \approx 2.6130$. Daily: $(1 + \frac{1}{365})^{365} \approx 2.7146$. Every second? The values creep up but never pass a certain number, $e = 2.71828\ldots$, the limit of $\left(1 + \frac{1}{n}\right)^n$. This number e is as important as π; you will meet it properly in stage C.

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8. Changing the time unit

If the growth factor per day is g_{day}, then a week is 7 days of multiplying, so

$$g_{\text{week}} = g_{\text{day}}^7.$$

The other way: an hour is $\frac{1}{24}$ of a day, and 24 hours must multiply up to one day: $g_{\text{hour}}^{24} = g_{\text{day}}$. Fractional exponents (§5) give the answer:

$$g_{\text{hour}} = g_{\text{day}}^{1/24}.$$

Worked example 8. A plant grows 10% per day, so $g_{\text{day}} = 1.1$.

- $g_{\text{week}} = 1.1^7 \approx 1.9487$, about $+94.9\%$ per week (not 70%).
- $g_{\text{hour}} = 1.1^{1/24} \approx 1.003979$, about $+0.398\%$ per hour (not $10/24 \approx 0.417\%$).

Keep at least four decimals in a growth factor; rounding 1.003979 to 1.00 throws away all the growth.

General rule. If g is the growth factor per time unit, then for a period of k time units the growth factor is g^k, for every positive k, whole or fractional.

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9. Doubling time and half-life

The doubling time (*verdubbelingstijd*) is the time it takes for an exponentially growing quantity to double. The half-life (*halveringstijd*) is the time it takes for a decaying quantity to halve.

The doubling time does not depend on the start value or on when you start. Starting at any moment t_0, the amount multiplies by g^T in the next T time units. It doubles exactly when $g^T = 2$, and that equation does not contain b or t_0.

So to find a doubling time, solve $g^T = 2$; for a half-life, solve $g^T = \frac{1}{2}$. Without logarithms we do this by **trial and improvement** with a table, or by reading the graph.

Worked example 9. Growth of 15% per year: solve $1.15^T = 2$.

T	4	4.9	5.0
1.15^T	1.7490	1.9834	2.0114

So T lies between 4.9 and 5.0. Since 2.0114 is closer to 2 than 1.9834, $T \approx 5.0$ years. (In whole years: after 5 years the amount has more than doubled.)

Using a known half-life. If a medicine has a half-life of 6 hours, then after t hours the fraction left is $\left(\frac{1}{2}\right)^{t/6}$, because $t/6$ half-lives have passed. The growth factor per hour is $g = 0.5^{1/6} \approx 0.8909$.

In C2 you will learn logarithms (*logaritmen*), which solve $g^T = 2$ exactly in one step: $T = \log_g 2$.

> **Going further: the rule of 72** For an interest rate of $p\%$ per period, the doubling time is roughly $72/p$ periods. At 6% per year: $72/6 = 12$ years, and indeed $1.06^{12} \approx 2.012$. At 8%: $72/8 = 9$ years, and $1.08^9 \approx 1.999$. Why 72? The exact answer needs logarithms (the true constant is near $100 \cdot 0.693 = 69.3$ for small p), but 72 has many divisors (2, 3, 4, 6, 8, 9, 12) and happens to work best for rates between about 4% and 12%. Problem 28 lets you test it.

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10. Linear or exponential?

Linear	Exponential
$y = ax + b$	$N = b \cdot g^t$
equal steps in x give a constant **difference** in y	equal steps in t give a constant **ratio** in N
graph: straight line	graph: curve, gets steeper and steeper (or flattens towards 0)
adds the same amount each step	multiplies by the same factor each step

Why a constant ratio means exponential: if each step multiplies by the same number g, then after t steps you have multiplied by g exactly t times, so $N = b \cdot g^t$. Conversely, for $N = bg^t$ the ratio of consecutive values is $\dfrac{bg^{t+1}}{bg^t} = g$, the same every time. So "constant ratio" and "exponential" are two descriptions of the same thing (for equally spaced t).

Worked example 10.

t	0	1	2	3
N	80	120	180	270

Differences: 40, 60, 90, not constant. Ratios: $\frac{120}{80} = 1.5$, $\frac{180}{120} = 1.5$, $\frac{270}{180} = 1.5$, constant. So $N = 80 \cdot 1.5^t$.

Real data are rarely exact: if the ratios are all close to the same number, the relation is *approximately* exponential and you use the average ratio.

Exponential always wins. However steep a linear function is, and however small $g > 1$ is, $b \cdot g^t$ eventually overtakes it. That is why exponential growth in nature (bacteria, viruses) never lasts long: something runs out.

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11. Power functions

A power function (*machtsfunctie*) has the form $y = a \cdot x^n$. Do not confuse it with an exponential function: in x^3 the **variable is the base**; in 3^x the **variable is the exponent**.

Even powers, $y = x^2$ and $y = x^4$. Because $(-x)^2 = x^2$, the points (x, y) and $(-x, y)$ are both on the graph: the graph is **symmetric in the y-axis** (*lijnsymmetrisch in de y-as*). Both are U-shaped with minimum $(0, 0)$ and go through $(-1, 1)$ and $(1, 1)$. $y = x^4$ is flatter near 0 (because $0.5^4 = 0.0625 < 0.25 = 0.5^2$) and steeper beyond $x = 1$.

Odd powers, $y = x^3$. Because $(-x)^3 = -x^3$, if (x, y) is on the graph then so is $(-x, -y)$: the graph is **point-symmetric in the origin** (*puntsymmetrisch in de oorsprong*). It rises everywhere and goes through $(-1, -1)$, $(0, 0)$, $(1, 1)$.

Negative powers, $y = x^{-1} = \frac{1}{x}$. This graph is a hyperbola (*hyperbool*). It has two branches, in the first and third quadrants, and it is point-symmetric in the origin (odd power). Why does it have asymptotes (*asymptoten*)?

- As x gets very large, $\frac{1}{x}$ gets very close to 0 but never reaches it: $\frac{1}{1000} = 0.001$, $\frac{1}{10^6} = 10^{-6}$. The graph approaches the line $y = 0$: a **horizontal asymptote**.

- As x gets close to 0 (from the positive side), $\frac{1}{x}$ gets huge: $\frac{1}{0.001} = 1000$. And $x = 0$ itself is not allowed (you cannot divide by 0). The graph approaches the line $x = 0$: a **vertical asymptote**.

An asymptote is a line that the graph approaches more and more closely, without end. (The precise definition comes with limits in stage C.)

$y = x^{-2} = \frac{1}{x^2}$ has the same two asymptotes, but both branches lie above the x-axis (the values are squares, so positive), and the graph is symmetric in the y-axis (even power).

$y = x^{1/2} = \sqrt{x}$ exists only for $x \geq 0$. It starts at $(0, 0)$ and rises ever more slowly. It is the top half of the parabola $y = x^2$ reflected in the line $y = x$, because squaring and taking roots undo each other.

Function	Domain	Symmetry	Asymptotes
$y = x^2$, $y = x^4$	all x	y-axis	none
$y = x^3$	all x	origin	none
$y = x^{-1}$	$x \neq 0$	origin	$x = 0$ and $y = 0$
$y = x^{-2}$	$x \neq 0$	y-axis	$x = 0$ and $y = 0$
$y = \sqrt{x}$	$x \geq 0$	none	none

The factor a in $y = ax^n$ stretches the graph vertically; a negative a reflects it in the x-axis.

Inverse proportionality (*omgekeerd evenredig*). Two quantities are inversely proportional when their **product** is constant: $x \cdot y = c$, so $y = \frac{c}{x}$. Doubling x halves y. Example: 12 workers need 10 days, so the job is $12 \cdot 10 = 120$ worker-days; 15 workers need $120/15 = 8$ days. Compare **directly proportional** (*recht evenredig*): $y = cx$, the quotient y/x is constant.

Worked example 11. The intensity of light from a lamp is $I = \dfrac{400}{d^2}$ (d in metres). At $d = 2$: $I = 100$. Doubling the distance to $d = 4$ gives $I = 25$: a quarter. In general multiplying d by k divides I by k^2, because $(kd)^2 = k^2 d^2$ (rule 4).

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12. Periodic phenomena

A relation is periodic (*periodiek*) when the graph repeats itself after a fixed time. Examples: the height of the water at the coast (tides, *eb en vloed*), the height of a seat on a Ferris wheel (*reuzenrad*), the pressure in a heartbeat on an ECG, the temperature over a year.

For a periodic graph that goes up and down between a fixed maximum and minimum:

- **period** (*periode*): the length of one full repeat, for example from one top to the next;
- **equilibrium line** (*evenwichtsstand*): the horizontal line halfway between the maximum and the minimum, $\dfrac{\max + \min}{2}$;
- **amplitude** (*amplitude*): the distance from the equilibrium line to the maximum, $\dfrac{\max - \min}{2}$.

Worked example 12 (Ferris wheel). A Ferris wheel has diameter 50 m, its lowest point is 3 m above the ground, and one turn takes 6 minutes. The seat height goes between 3 m and 53 m. Equilibrium line: $\frac{53+3}{2} = 28$ m (the height of the axle). Amplitude: $\frac{53-3}{2} = 25$ m (the radius). Period: 6 minutes. If the seat is at the bottom at $t = 0$, it is at the top at $t = 3$, back at the bottom at $t = 6$, and after 20 minutes it is in the same place as after $20 - 3 \cdot 6 = 2$ minutes.

Using the period. To find the value at a late time, subtract whole periods. To find the next maximum, add one period to a known maximum.

Heartbeat. An ECG graph is periodic but not a smooth wave: a sharp spike, then small bumps. The period is the time between spikes; with a period of 0.8 s the heart beats $60/0.8 = 75$ times per minute. Period and frequency (repeats per time unit) are each other's reciprocal: an example of inverse proportionality from §11.

The smooth, wave-shaped periodic graphs of tides and Ferris wheels are all stretched and shifted versions of one graph, the sine graph (*sinusoïde*). You will study it, with formulas like $h = 28 - 25\cos\left(\frac{2\pi}{6}t\right)$, in C4 (trigonometric functions).

Summary

- Rules: $a^p a^q = a^{p+q}$, $a^p/a^q = a^{p-q}$, $(a^p)^q = a^{pq}$, $(ab)^p = a^p b^p$.
- Forced by the rules: $a^0 = 1$, $a^{-n} = 1/a^n$, $a^{1/n} = \sqrt[n]{a}$, $a^{p/q} = (\sqrt[q]{a})^p$.
- $\sqrt{a}\sqrt{b} = \sqrt{ab}$ for $a, b \geq 0$; there is no rule for $\sqrt{a+b}$.
- $\sqrt{2}$ is irrational (proof by contradiction).
- $N = b \cdot g^t$; percentage p per period gives $g = 1 \pm p/100$; percentages multiply, they do not add.
- Change time unit with powers: $g_{\text{week}} = g_{\text{day}}^7$, $g_{\text{hour}} = g_{\text{day}}^{1/24}$.
- Doubling time: solve $g^T = 2$; half-life: solve $g^T = \frac{1}{2}$ (by table now, by logarithms in C2).
- Linear: constant difference. Exponential: constant ratio.
- Even powers: symmetric in the y-axis. Odd powers: symmetric in the origin. x^{-1} and x^{-2} have asymptotes $x = 0$ and $y = 0$.
- Periodic: period, amplitude $= \frac{\max - \min}{2}$, equilibrium line $= \frac{\max + \min}{2}$.